

CLOUD COMPUTING PRACTICAL VIVA

Simple 2-Mark Questions and Answers - Easy English for Viva Preparation

Tip: For a 2-mark viva, remember the first sentence of each answer. Add the second sentence only when the examiner asks for more detail.

Experiment 1: Amazon EC2 and AWS Services

Q. What is AWS?

Ans. AWS stands for Amazon Web Services. It provides cloud services like computing, storage, databases and networking.

Q. What is Amazon EC2?

Ans. EC2 stands for Elastic Compute Cloud. It provides virtual servers in the cloud to run applications.

Q. Why is EC2 called elastic?

Ans. Because computing resources can be increased or decreased according to demand.

Q. What is an EC2 instance?

Ans. An EC2 instance is a virtual machine running on AWS cloud.

Q. What is Amazon S3?

Ans. S3 is a cloud storage service used to store files, images, videos and other data.

Q. Difference between EC2 and S3?

Ans. EC2 is mainly used for computing, while S3 is used for storing data.

Q. What is an AMI?

Ans. AMI stands for Amazon Machine Image. It is a template used to create EC2 instances.

Q. What is a cloud region?

Ans. A region is a geographical area where AWS has data centers.

Experiment 2: VirtualBox / VMware

Q. What is virtualization?

Ans. Virtualization is the process of creating virtual computers or resources using physical hardware.

Q. What is a virtual machine?

Ans. A virtual machine is a software-based computer that runs an operating system on virtual hardware.

Q. What is VirtualBox?

Ans. VirtualBox is virtualization software used to create and run virtual machines.

Q. What is VMware?

Ans. VMware is software that allows multiple operating systems to run on one physical machine using virtualization.

Q. What is a host OS?

Ans. The host OS is the operating system installed directly on the physical computer.

Q. What is a guest OS?

Ans. The guest OS is the operating system running inside a virtual machine.

Q. What is a hypervisor?

Ans. A hypervisor is software that creates and manages virtual machines.

Q. Give one advantage of virtualization.

Ans. It allows multiple operating systems to run on one physical machine and improves hardware utilization.

Experiment 3: Google App Engine

Q. What is Google App Engine?

Ans. Google App Engine is a Platform as a Service (PaaS) used to deploy applications on Google Cloud.

Q. Why is App Engine called PaaS?

Ans. Because it provides a platform to develop and deploy applications without managing physical servers.

Q. Which applications can run on App Engine?

Ans. Web applications and APIs written in supported programming languages can run on App Engine.

Q. What is the main advantage of App Engine?

Ans. It manages servers for us and can scale applications according to traffic.

Q. What is automatic scaling?

Ans. It means increasing or decreasing resources automatically based on application demand.

Q. What is deployment?

Ans. Deployment means uploading an application and making it available to users on the cloud.

Experiment 4: Google App Engine Launcher

Q. What is Google App Engine Launcher?

Ans. It is an older GUI-based tool used to create, run and deploy Google App Engine applications.

Q. Why is a launcher used?

Ans. It makes application management easier by providing a graphical interface.

Q. What can we do using the launcher?

Ans. We can create, run locally, test and deploy App Engine applications.

Q. What is local testing?

Ans. Local testing means running the application on our own computer before cloud deployment.

Q. What is cloud deployment?

Ans. It means making the application run on cloud infrastructure so users can access it online.

Experiment 5: CloudSim - Load Balancing

Q. What is CloudSim?

Ans. CloudSim is a simulation toolkit used to model and test cloud computing environments.

Q. Why is CloudSim used?

Ans. It helps researchers test cloud algorithms without using real cloud infrastructure.

Q. What is load balancing?

Ans. Load balancing distributes workload among multiple resources so that no resource becomes overloaded.

Q. Why is load balancing important?

Ans. It improves performance, resource utilization and response time.

Q. What is a virtual machine in CloudSim?

Ans. A VM is a simulated computing resource on which cloud tasks can be executed.

Q. What is a data center in CloudSim?

Ans. A data center represents the physical infrastructure that provides computing resources.

Q. What is a broker in CloudSim?

Ans. A broker manages users requests and sends tasks to suitable virtual machines.

Q. What is simulation?

Ans. Simulation means creating a model of a real system and testing its behavior without using the actual system.

Experiment 6: Transfer Files Between Virtual Machines

Q. Why do we transfer files between VMs?

Ans. To share data, applications or configuration files between different virtual machines.

Q. What is a shared folder?

Ans. A shared folder is a directory that can be accessed by both the host and virtual machine.

Q. What is SCP?

Ans. SCP stands for Secure Copy Protocol. It is used to securely transfer files between computers.

Q. What is FTP?

Ans. FTP stands for File Transfer Protocol. It is used to transfer files over a network.

Q. What is SSH?

Ans. SSH stands for Secure Shell. It provides secure remote access to another computer.

Q. Main requirement for transferring files between VMs?

Ans. The VMs should have network connectivity and proper permissions.

Experiment 7: OpenStack using TryStack

Q. What is OpenStack?

Ans. OpenStack is an open-source cloud platform used to manage compute, storage and networking resources.

Q. What is TryStack?

Ans. TryStack is an environment that allows users to experiment with OpenStack services.

Q. What is OpenStack used for?

Ans. It is used to create and manage virtual machines, networks and cloud resources.

Q. What is Nova?

Ans. Nova is the OpenStack compute service used to manage virtual machines.

Q. What is Neutron?

Ans. Neutron is the OpenStack networking service used to create and manage networks.

Q. What is a cloud instance in OpenStack?

Ans. An instance is a virtual machine created and running in the OpenStack cloud.

Q. What is an image in OpenStack?

Ans. An image is a template containing an operating system used to create virtual machines.

Q. Benefit of OpenStack?

Ans. It provides an open-source platform for building and managing private or public clouds.

Experiment 8: Hadoop Single Node Cluster

Q. What is Hadoop?

Ans. Hadoop is an open-source framework used to store and process large amounts of data.

Q. What is HDFS?

Ans. HDFS stands for Hadoop Distributed File System. It stores large files in distributed form.

Q. What is MapReduce?

Ans. MapReduce is a programming model used to process large datasets in parallel.

Q. What is a single-node cluster?

Ans. A single-node cluster runs Hadoop services on one machine.

Q. What is NameNode?

Ans. NameNode manages the metadata and structure of files stored in HDFS.

Q. What is DataNode?

Ans. DataNode stores the actual data blocks in HDFS.

Q. Advantage of Hadoop?

Ans. Hadoop can process large datasets using distributed storage and parallel processing.

Q. What is big data?

Ans. Big data means very large and complex datasets that are difficult to process using traditional systems.

Experiment 9: IPTV Service Deployment Architecture

Q. What is IPTV?

Ans. IPTV stands for Internet Protocol Television. It delivers television content using IP networks.

Q. What is video streaming?

Ans. Video streaming allows users to watch video while it is being delivered over a network.

Q. What is CDN?

Ans. CDN stands for Content Delivery Network. It uses distributed servers to deliver content faster.

Q. Why is CDN used in IPTV?

Ans. It reduces delay and improves video delivery by serving content from servers closer to users.

Q. What is load balancing in IPTV?

Ans. It distributes user requests and video traffic across multiple servers.

Q. Why is load balancing needed?

Ans. It prevents server overload and improves system performance and availability.

Q. What is a video streaming server?

Ans. It stores, processes and delivers video content to users.

Q. What happens if one server becomes overloaded?

Ans. Load balancing can redirect new requests to other available servers.

Experiment 10A: Healthcare Data Analytics on Cloud

Q. What is healthcare data analytics?

Ans. It is the process of analyzing healthcare data to find useful patterns and support decisions.

Q. Why is cloud used in healthcare analytics?

Ans. Cloud provides scalable storage and computing power for processing large healthcare datasets.

Q. Give examples of healthcare data.

Ans. Patient records, medical reports, test results and hospital data are examples.

Q. What is cloud storage?

Ans. Cloud storage means storing data on remote servers that can be accessed through the Internet.

Q. What is scalability?

Ans. Scalability is the ability of a system to increase or decrease resources according to workload.

Q. One advantage of cloud analytics?

Ans. It provides flexible computing resources and reduces the need for expensive local hardware.

Experiment 10B: Satellite Image Analytics on Cloud

Q. What is satellite image analysis?

Ans. It is the process of analyzing satellite images to extract useful information.

Q. Why is cloud useful for satellite image processing?

Ans. Satellite images are large, so cloud computing provides high storage and processing power.

Q. What is image classification?

Ans. Image classification means assigning an image or its parts to predefined categories.

Q. What is image preprocessing?

Ans. It is the process of preparing or improving an image before analysis.

Q. Applications of satellite image analytics?

Ans. Agriculture, weather monitoring, disaster management and land-use analysis.

Q. What are cloud storage and cloud compute?

Ans. Cloud storage stores large image datasets, while cloud compute processes and analyzes them.

Most Important Common Viva Questions

Q. What is cloud computing?

Ans. Cloud computing means using computing resources like servers, storage and software over the Internet.

Q. What is IaaS?

Ans. IaaS provides infrastructure like virtual machines, storage and networking over the Internet.

Q. What is PaaS?

Ans. PaaS provides a platform for developing and deploying applications without managing the underlying infrastructure.

Q. What is SaaS?

Ans. SaaS provides software applications over the Internet, usually through a web browser.

Q. What is scalability?

Ans. Scalability means increasing or decreasing resources according to workload.

Q. What is elasticity?

Ans. Elasticity means automatically adding or removing resources when demand changes.

Q. What is load balancing?

Ans. Load balancing distributes work among multiple servers or resources.

Q. What is a virtual machine?

Ans. A VM is a software-based computer that runs on virtual hardware.

Q. What is a hypervisor?

Ans. A hypervisor is software that creates and manages virtual machines.

Q. What is AWS?

Ans. AWS is Amazon's cloud platform that provides services such as EC2, S3, databases and networking.

Q. What is Hadoop?

Ans. Hadoop is a framework for distributed storage and processing of big data.

Q. What is HDFS?

Ans. HDFS is Hadoop's distributed file system used to store large datasets.

Q. What is CloudSim?

Ans. CloudSim is a simulator used to model and test cloud computing systems and algorithms.

Q. What is CDN?

Ans. CDN is a network of distributed servers used to deliver content faster to users.

Quick Memory Sheet

Cloud: Computing resources over the Internet.

VM: Virtual computer.

Hypervisor: Software that manages VMs.

AWS: Amazon cloud platform.

EC2: Cloud virtual server.

S3: Cloud storage.

CloudSim: Cloud simulation tool.

OpenStack: Open-source cloud platform.

Hadoop: Big-data processing framework.

HDFS: Hadoop distributed storage.

CDN: Distributed servers for fast content delivery.

Load Balancing: Distributing workload among servers.

Scalability: Increasing or decreasing resources as needed.

Elasticity: Automatic resource adjustment according to demand.